and the AWT components. Developers were forced to test their applications on each platform, a practice that became infamously known as write once, debug everywhere.

A year after Java 1.0 came out, Netscape created a Java GUI library called the IFC (Internet Foundation Classes) that used a fresh approach. The IFC was a set of lightweight classes based on ideas from NEXTSTEP's User Interface Toolkits. These did not rely on peers. Javasoft, fearing that the Java community would fracture because of the new IFC, made a deal with Netscape to adapt the IFC. Using the IFC approach, User Interface Components (buttons, menus, textboxes—called Widgets) were painted onto blank windows. The only peer functionality needed was a way to put up windows and paint on the window. Netscape's IFC widgets looked and behaved the same no matter which platform they ran on.

Sun accepted Netscape's lead on this process and created a new user interface library named after a Duke Ellington song:

"It Don't Mean A Thing If It Ain't Got That Swing..."

Swing is now the official name for the non-peer-based GUI toolkit that is part of the Java Foundation Classes (JFC). Because Swing was not part of the original core Classes, but rather, was an extension, it was named javax.swing. Thus, the AWT library is still used, especially the version 1.1 Event Handling library. Swing elements are somewhat slower than their AWT counterparts, so the AWT elements are still valuable. The AWT library of classes is still needed to this day—Swing is built on AWT. But Swing takes advantage of the AWT's infrastructure, including graphics, colors, fonts toolkits and layout managers.

Now you need to remember that Swing does NOT use the AWT's components. Of all the components the AWT has to offer, Swing uses only Frame, Window and Dialog, as the Superclasses to JFrame, JWindow and JDialog. In effect, Swing cherry picked the best of the AWT to build a new set of mostly lightweight components, while discarding the parts of AWT that were the most trouble—the heavyweight components. It was intended to replace the AWT's heavyweight components, but not the AWT itself. So, in order to understand the Swing GUI library (next week), we need to understand the AWT library first.